Project 1 - Group 18

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November 14, 2025

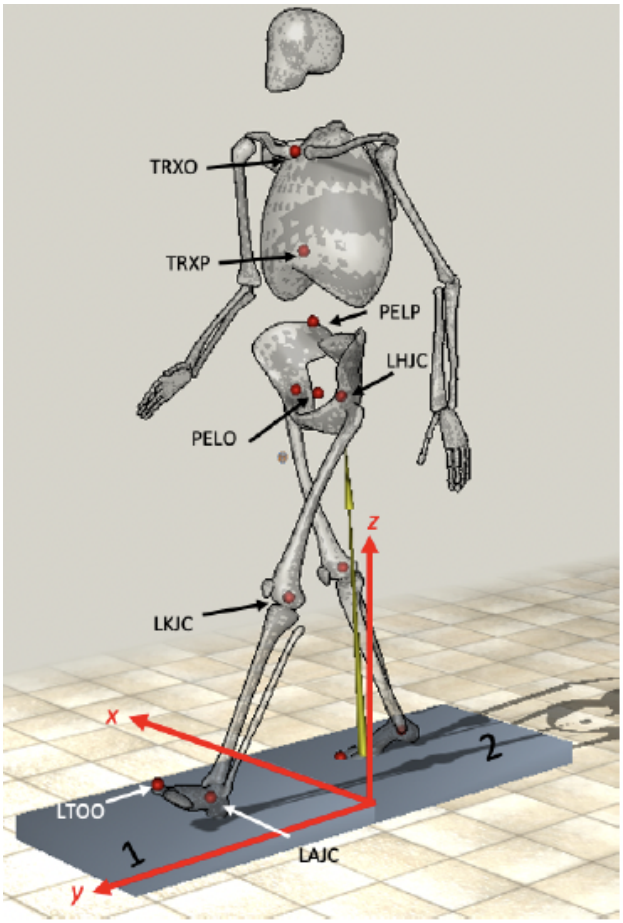


Figure 1: Model showing coordinates and centre outputs.

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1 Introduction

In this project, we utilized laboratory-measured motion capture data and force plate data to calculate and plot multiple joint angles in the sagittal plane—including the trunk, pelvis, hip, knee, and ankle—as the subject performed walking and jogging. Subsequently, based on anthropometric assumptions, we employed the inverse dynamics method to calculate the internal moments and joint power at the ankle, knee, and hip joints.

Through the quantitative analysis of these parameters during movement, we gained an intuitive understanding of the body's posture and loads under different gait patterns. This helps us comprehend the function of muscles and the skeleton during motion, such as supporting the body and facilitating power generation and absorption, thereby providing insight into the entire gait cycle.

Contribution

All group members collaborated on the MATLAB code simultaneously during Zoom meetings. When minor adjustments or small tweaks where made individually, they were presented at the next meeting or communicated through other channels.
Ioanna Eleni Leonidou – Method and Results section.
Sahand Khodabandehloo – Discussion section.
Liewen Si – Introduction and Discussion section.

2 Method

2.1 Motion capture

The experimental data was collected at the KTH MoveAbility Laboratory. The protocol utilized a 3D motion capture system, which consisted of multiple infrared cameras surrounding the force plates embedded in the floor and IMU (Inertial Measurement Units) sensors position strategically on the subject.

The camera sampling frequency was 100 Hz, used to track reflective markers on the subject to obtain kinematic data. The force plates recorded ground reaction forces (GRF) at an original sampling frequency of 1000 Hz, which was subsequently down-sampled to 100 Hz for synchronization with the motion capture data, enabling the kinematic and force data to be combined for the inverse dynamics calculations.

During data collection, the subject was instructed to perform different gait tasks, including "Normal walking", "Jogging", and "Crouch walking". To acquire kinetic data for a specific gait cycle, the trial was structured so the subject's foot would land on a designated force plate. The subject's starting position was adjusted by the experimenters as needed to ensure the gait pattern was as natural as possible.

The gait cycle was defined as the temporal period from initial heel-strike to the subsequent heel-strike of the same foot, and all data were normalized to 0-100% of the gait cycle for comparative analysis. Joint angles were calculated using absolute and relative angles such that hip angle is femur relative to pelvis, knee angle is tibia relative to femur and ankle angle is foot relative to tibia, all angles are reported in degrees. Joint moments and power were normalized to the subjects body mass (kg), resulting in units of Nm/kg and W/kg respectively.

During walking the right foot contacted force plate 2 and the left foot contacted force plate 3. The analysed gait cycles (heel strike to heel strike) were for the right leg frames 562 - 659 and for the left frames 511 - 609. For jogging only the left leg was analysed which contacted force plate 2. Frames 166 to 236 where analysed for jogging.

All coordinates were expressed in the sagittal y–z plane (horizontal–vertical) and converted from millimetres to meters before analysis. MATLAB® R2025a was used to analyse and present data.

2.2 Kinematic Computations

Segment angles were computed from the marker coordinates using the signed angle between two segment vectors. The atan2 formulation ensures correct sign and continuity.

$\theta = \text{atan2}(u_y v_z - u_z v_y,\, u_y v_y + u_z v_z)$

where u and v are 2D vectors defining consecutive segments.

All angles were reported in degrees, but time derivatives (angular velocities ω and accelerations α were computed in radians, as required for inverse dynamics.

Trunk and pelvis angles were defined as absolute angles relative to the vertical axis y. Whereas hip angle was defines as thigh relative to pelvis, knee as shank relative to thigh and ankle foot relative to shank with an offset of 4 degrees of dorsiflexion.

2.3 Kinetic Computations

The inverse dynamics calculations were implemented in a frame-by-frame loop, proceeding distal to proximal (ankle - knee - hip). At each time frame, the positions of the ankle, knee, hip, segment centre of mass (COM), ground reaction force (GRF) and centre of pressure (COP) were used on the Newton–Euler equations for a planar rigid body2.

The computations followed the free-body diagrams as explained and shown in detailed below.

For the moment of inertia we used the segment mass times the radius of gyration squared [see Figure 3]. As the radius of gyration was divided by length in the table 4.1 of Winter (2009) the length of each segment was also calculated, multiplied and squared to assure representative values.

Finally for the power, the simplified power equation for the sagittal plane was used as shown in Figure 4.

$$\sum \mathbf{M}_p = \frac{d\mathbf{H}_p}{dt} = \sum \mathbf{r}_{pA} \times \mathbf{F} + \mathbf{M}_{ext} = I_p \boldsymbol{\alpha}$$

Figure 2: Rate of change of angular momentum H about a point p that is fixed in an inertial reference frame.

$$I_{Gj} = m_j \rho_{Gj}^2$$

Figure 3: Moment of inertia I for segment j where ρ is the radius of gyration.

Joint power was calculated as:

$$P_j = M_{j,x} \cdot \omega_{j,x}$$

Figure 4: Simplified power equation for the sagittal plane.

2.3.1 Ankle joint

In the kinetics part the foot was modeled as having negligible mass and inertia, a simplification accepted in biomechanics literature [1]. Therefore the ankle reaction force was set equal to the negative ground reaction force (GRF). The ankle moment was then computed from the GRF and COP using a 2D cross product as

per the below free-body diagram [see Figure 5.

$$Mankle = -[rCOP - rA) \times FGRF + Mground$$

Where GRF acts at the COP, the moment arm is the difference we get from COP - ankle and the external moment on the ground (force plate) is added [see Figure 5].

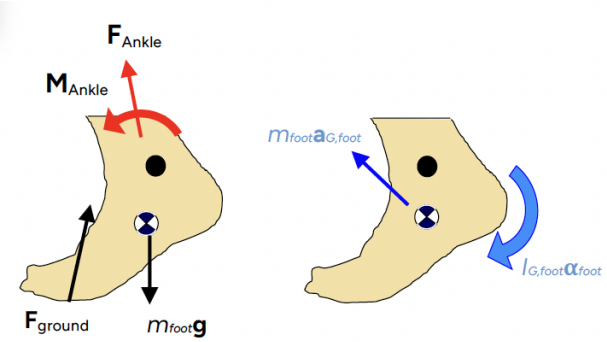


Figure 5: Free body diagram for ankle. Picture obtained from Lecture 4 slides in course material.

2.3.2 Knee joint

To calculate the moment about the knee joint we used the moment caused by the distal force at the ankle as well as the distal ankle moment opposite direction to the knee moment which were calculated as per section 2.3.1. In addition to the ankle force and moment we also took into account the gravitational torque about the knee, as well as the linear inertial torque and the rotational inertia of the shank as illustrated in the right shank with blue arrows. These values were inserted in the Newton-Euler equation [see Figure 2].

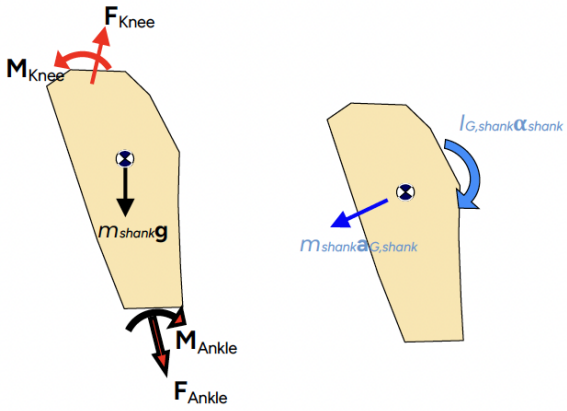


Figure 6: Free body diagram for knee. Picture obtained from Lecture 4 slides on the course material.

2.3.3 Hip joint

For the hip joint calculations the reaction forces and moments calculated from the knee joint where used as distal loads which where inserted in the Newton-Euler formula [see Figure 2].
Detailed forces and moments are shown in the below free body diagram for the hip joint [Figure 7].

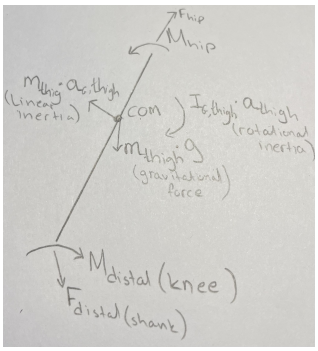


Figure 7: Hand drawn sketch of hip free-body diagram as used in the calculations.

2.3.4 Left leg and jogging

The left leg code mirrors the right leg exactly, using the respective left leg values. For the jogging analysis we simply adjusted the input data path and frames to account for the corresponding jogging data.

2.4 Anthropometry

Anthropometric properties were derived from Winter (2009, Table 4.1), using the subject's height (1.67 m) and body mass (70.7 kg). Segment parameters used:

Segment	Mass(%BW)	COM(%from proximal)	Radius of gyration(ρ)
Foot	1.37	50	0.475
Shank	4.65	43.3	0.302
Thigh	10.0	43.3	0.323
Pelvis	14.2	25	-

Table 1: Anthropometric properties used, derived from Winter (2009, table 4.1)[2]

3 Results

The following figures present the computed joint kinematics and kinetics for the walking and jogging data. Joint angles are presented in degrees, while their time derivatives (angular velocity and acceleration) moments, and powers were computed in radians. Right leg results are shown in red and left leg results in blue. Each plot represents a full gait cycle per leg.

3.1 Normal walking

During the gait cycle, both limbs show similar angular patterns, confirming approximately symmetric gait.

3.1.1 Sagittal Plane Angles

Hip flexes during swing (60-100% of gate cycle) and extends in stance (0-60% of the gate cycle). Knee flexes at heel strike and again during swing. Ankle dorsiglexes during stance then flat during midstance and plantarflexes just before toe-off.

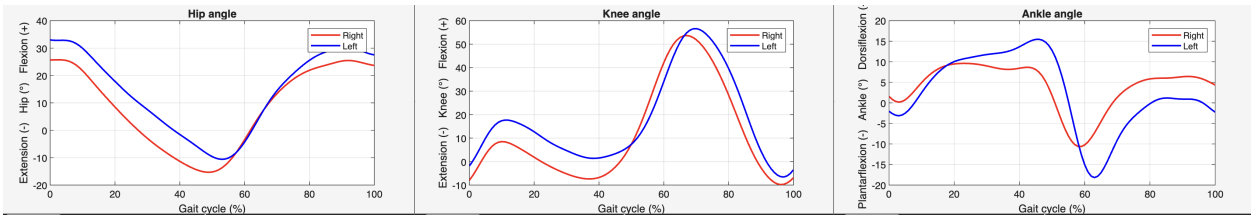


Figure 8: Ankle, knee and hip angles in the sagittal plane for normal walking.

3.1.2 Joint Moments

For the ankle moment arm is largest just before the toe-off at about 50% of the gait cycle as the bodys center of gravity is furthest from the ankle just before toe-off. The knee there is an extensor moment in early stance and again just before toe-off. For the hip there is a extensor moment in early stance just at heel strike which then flexes until about 50% of the gait cycle when it again flexes.

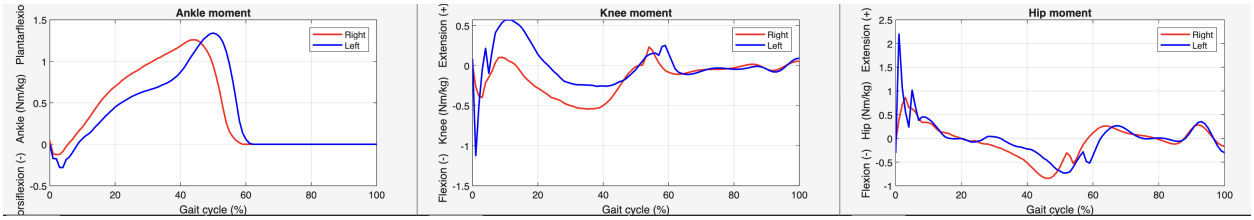


Figure 9: Ankle, knee and hip moments in the sagittal plane for normal walking.

3.1.3 Joint Power

For the ankle there is a major power generation just before toe-off during push off. The knee has a more alternating power absorption with smaller generations but mainly at around toe-off. Hip is generating power during most of the stance until just before toe-off when there is a large drop before it again generates power just after toe-off. The ankle joint plot shows a larger power range compared to the knee and hip joint.

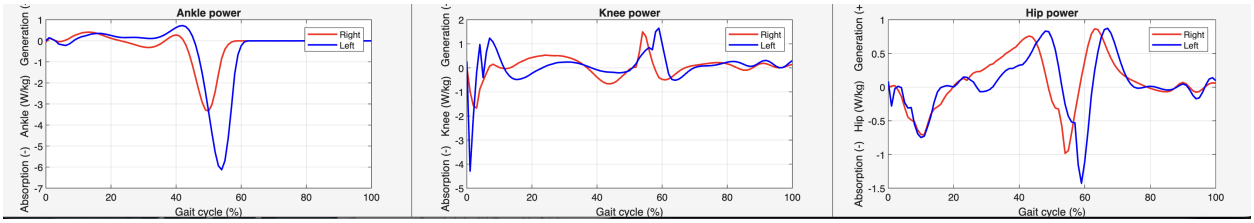


Figure 10: Ankle, knee and hip power in the sagittal plane for normal walking.

3.2 Normal walking vs. Jogging

Figure 11 shows the sagittal plane joint angles for the left leg during the selected jogging cycle (frames 166 to 236), normalized to 0–100% gait. Compared to walking, peak hip and knee flexion are larger and occur earlier in stance, while ankle dorsiflexion at initial contact is greater.

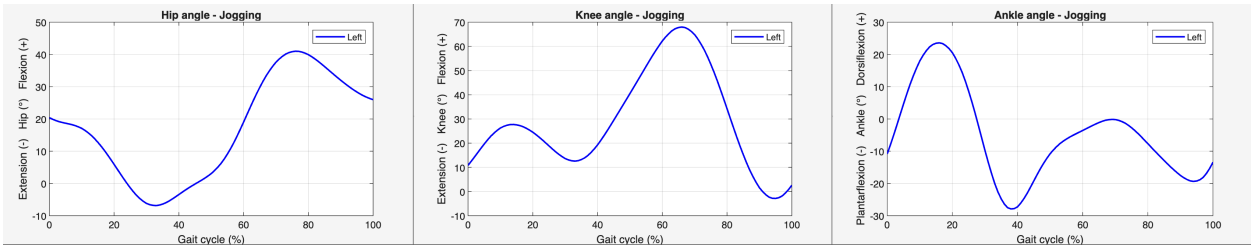


Figure 11: Ankle, knee and hip angles in the sagittal plane for jogging.

The computed joint moments [see Figure 12] similarly to the angles show increased magnitudes during jogging compared to normal walking. Peak knee extensor moment reached approximately 1 Nm/kg near early stance and peak ankle plantarflexion moment reached 3 Nm/kg. Hip moment was however lower compared to normal walking with peak extension at 0.6 Nm/kg compared to over 2 Nm/kg at normal walking.



Figure 12: Ankle, knee and hip moments in the sagittal plane for jogging.

Joint power, see Figure 13, show greater values for ankle with peak absorption at 15 Nm/kg compared to 6 Nm/kg for normal walking. Knee power has no significant power difference whereas the hip shows an increased power generation during stance reaching peak at 3 Nm/kg.

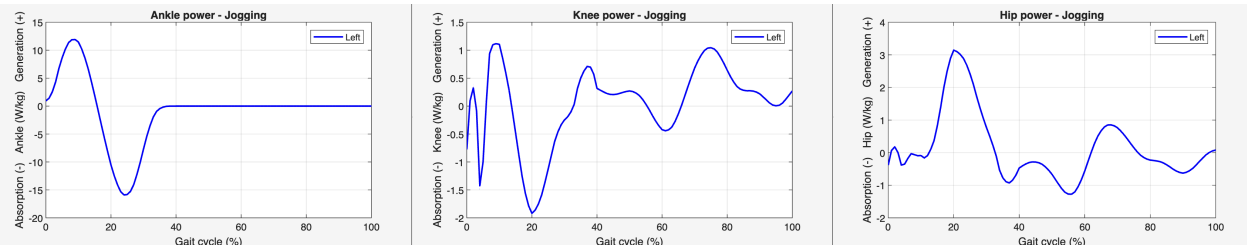


Figure 13: Ankle, knee and hip power in the sagittal plane for jogging.

4 Discussion

In the previous sections, we obtained quantitative data for the subject during normal walking and jogging. This data can serve as the basis for gait analysis. For example, power and moments can reveal muscle group activation, while the differences between the curves for the left and right legs can help us analyze the symmetry of the subject's movement.

4.1 Normal walking

According to the instructions, hip extensor moment, knee extensor moment and ankle plantarflexor are defined as positive. This means that when you try to extend your leg and stand on your tiptoes, the muscles will produce a positive moment. It is worth mentioning that in the gait data obtained from the laboratory, the toe-off frame for the right leg during normal walking is frame 618 , and for the left leg is frame 571 ; the

toe-off frame for jogging is frame 193. These occur at approximately 60% of the normal walking gait cycle and 38% of the jogging gait cycle, respectively. This is the dividing point between the stance phase and the swing phase. Here, We will discuss the muscle recruitment and movement patterns of the three joints during the normal walking gait cycle.

Ankle Joint. The moment and power curves of the ankle joint indicate that the gait cycle can be broadly divided into four phases. First (Gait cycle 0% - 15%), the moment is negative, and power approaches zero. This indicates that the ankle's dorsiflexor muscles exert a slight force through an isometric contraction pattern, maintaining foot stability during this phase. Secondly (Gait cycle 15% - 40%), the moment becomes positive and rises steadily. The power becomes negative . This indicates that the plantarflexor muscles are active. Their activity mode is an eccentric contraction, as they are controlling the body lean forward. Thirdly (Gait cycle 40% - 60%), the moment reaches its maximum positive peak, indicating maximal activation of the plantarflexor muscles. However, the power curve shows a very large negative peak, signifying an absorption of energy. This means the muscles are performing an eccentric contraction. Finally (Gait cycle 60% - 100%), both the moment and power curves return to zero. This corresponds to the swing phase. It means the subject's toes completely lift off the ground, and the ankle muscles are largely relaxed as the leg swings forward to prepare for the next foot contact.

Knee Joint. Based on the curves of moment and power,the gait cycle is can also divided into 4. First (Gait cycle 0% - 20%), the moment is initially negative and then becomes strongly positive, while the power is predominantly negative. This indicates that the knee flexors first activate eccentrically, followed immediately by the knee extensors, which also work eccentrically to absorb shock and control the body's weight. Secondly (Gait cycle 20% - 50%), a positive moment is paired with positive power. This indicates the knee extensors are active in a concentric contraction to support the body's weight and actively extend the knee. Thirdly (Gait cycle 50% - 60%), just before toe-off, the moment reaches its second major positive peak, corresponding with a large positive power peak. This signifies a powerful concentric contraction of the knee extensors, generating energy for moving forward. Finally (Gait cycle 60% - 100%), during the swing phase, the moment becomes negative and the power also becomes negative. This indicates the knee flexors are performing an eccentric contraction to decelerate the swinging lower leg, preparing it for the next foot contact.

Hip Joint. The moment and power curves of the hip joint can be divided into the following phases. First (Gait cycle 0% - 15%), the moment is strongly positive, but the power is negative . This indicates that the hip extensor muscles are performing an eccentric contraction. They are acting as brakes to control the body's momentum and prevent the trunk from collapsing into flexion over the leg newly planted on the ground. Second (Gait cycle 15%-50%), still within the stance phase, the biomechanical pattern becomes complex. Negative torque and positive power coexist simultaneously. This indicates that the hip flexor muscles are undergoing concentric contraction, assisting the trunk in reaching forward over the supporting leg. Thirdly (Gait cycle 50% - 60%), the moment becomes strongly negative while the power shows a large negative peak. This is a clear sign that the hip flexor muscles are working eccentrically. They are stretching and absorbing energy to brake the rapid extension of the hip as the body moves past the foot. Finally (Gait cycle 60% - 100%), there is a positive power peak that corresponds with a positive moment peak. This indicates a powerful concentric contraction of the hip extensors which provides instantaneous power for forward motion. During the subsequent swing, the moment and power curves drop, but a small positive moment appears towards the end. This signifies an eccentric contraction of the hip extensors to decelerate the swinging leg in preparation for the next cycle.

4.2 Normal walking vs. Jogging

In contrast to normal walking, jogging (Figures 10-12) shows clear increases in joint excursions and mechanical demands at higher speeds. Jogging produces larger hip, knee, and ankle flexion angles, particularly the pronounced knee flexion during early stance. This increased flexion allows for greater shock absorption at landing.

4.2.1 Sagittal Plane Joint Angles (y-z-plane)

The plotted angles (Figure 8) show that jogging leads to noticeably larger joint excursions than walking, especially at the knee, where the flexion peak during early stance is much higher.

A significant difference is seen at the ankle. The total range of motion is almost doubled in jogging. This is achieved by both a greater peak dorsiflexion during mid-stance and a substantially larger peak plantarflexion at toe-off, which facilitates a much more powerful push-off.

During jogging, the hip joint has a greater range of motion. Although the initial flexion upon foot contact is lower than during normal walking, the peak flexion is significantly greater, exceeding 40°. This indicates that more hip extensor muscles are engaged to control trunk forward lean during jogging. The peak extension during jogging is approximately -10°, similar to that maintained during walking. This indicates that the hip joint does not have sufficient time to fully extend.

4.2.2 Joint Moments

The joint moments (Figure 11) also increase during jogging. The ankle plantarflexors work harder both to control the tibia as the body moves forward and to generate a stronger push-off. The quadriceps experience higher loads during early stance, where they absorb more energy as the knee flexes. The hip extensors also play a larger role, helping stabilize the trunk and control the faster forward motion of the thigh.

Ankle joint. For the ankle (left-hand side plot of Figure 11), the peak plantarflexor moment during jogging is about 3 Nm/kg, occurring in early stance (centered approximately between 0-35% of the gait cycle). This is an increase ($\approx 114\%$) from about 1.4 Nm/kg in walking, indicating substantially higher demands on the plantarflexors to control and generate push-off. Thus, during early-stance in jogging, the quadriceps act eccentrically to control the larger knee flexion and absorb the higher impact forces. Additionally, at the ankle, the plantarflexors first act eccentrically as the tibia moves forward over the foot, and then transition to a strong concentric contraction during push-off.

Knee joint. For the knee (central plot of Figure 11), the peak extensor moment during jogging is about 0.9 Nm/kg, occurring in early stance around 18% of the gait cycle. Compared with walking, this increase ($\approx 64\%$) reflects greater eccentric loading of the quadriceps to absorb impact and control the larger knee flexion seen in the angle plots.

Hip joint. For the hip (right-hand side plot of Figure 11), the peak extensor moment during jogging reaches about 0.65 Nm/kg at roughly 5% of the gait cycle. Although this is a decrease in early-stage comparison with normal walking ($\approx 71\%$), the fluctuation magnitudes of the mid- and end-stage remain roughly the same as previously, which implies that some of the early-stance loading is shifted away from the hip toward the knee and ankle. At the hip, the gluteus maximus and hamstrings also perform eccentric work immediately after foot strike to control forward trunk motion.

The kinematic and kinetic curves for the left and right sides during normal walking appear generally similar in shape, timing, and magnitude, indicating overall symmetric gait mechanics. Small side-to-side differences are expected due to natural biological variability, but no major asymmetries are visible in the plotted angles, moments, or powers. This suggests that the subject exhibits functionally symmetric lower-limb movement patterns and loading during the gait cycle.

4.2.3 Joint Powers

The power curves (Figure 13) reflect these changes. Jogging involves more energy absorption at the knee and hip during landing, and a much larger positive power burst at the ankle during push-off. Although the ankle remains the primary source of propulsion, the hip contributes more than it does during normal walking.

Ankle joint. In walking (Figure 10), power generation is minimal, with peaks barely reaching 0.5 W/kg. The stance phase is dominated by a large power absorption peak of -3 to -6 W/kg. In contrast, jogging (Figure 13) displays an explosive positive power burst of approximately 12 W/kg early in the stance phase (10% cycle). This propulsive burst is followed by an even larger absorption phase to control the movement before the foot leaves the ground.

Knee joint. The knee's role in energy management is heightened in jogging. Both gaits show an initial power absorption at landing. However, unlike walking, the knee exhibits a second peak power absorption at -2.0 W/kg during jogging. This indicates that the knee performs more eccentric work to manage the higher impact forces and control the greater flexion during the jogging stance.

Hip joint. During jogging, the hip's generation is massively increased. The main generation peak in jogging reaches approximately 3.5 W/kg , which is about 5 times greater than the peak generation seen during the walking stance (0.7 W/kg). This clearly shows the hip is a far more active power generator in jogging.

4.2.4 Analysis

Based on the discussed results, the same muscle groups are NOT equivalently activated during normal walking and jogging; it is concluded that the knee and ankle joints are responsible for more momentum and power generation during jogging compared to walking, where their relative contributions and workloads increase noticeably with the subject's speed, while the loading on the hip joint is decreased.

References

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[2] D. A. Winter, *Biomechanics and Motor Control of Human Movement*, 4th. Hoboken, New Jersey: John Wiley Sons, 2009, ISBN: 978-0-470-39818-0.